

Dhananjaya Paliwal

Denton, Texas 76201, USA
937-514-3573 | dhananjayapali12@gmail.com
[LinkedIn](#) | [GitHub](#) | [Medium](#)

Career Profile

Master's student in Data Science with hands-on experience in Generative AI, large language models, data analytics, and machine learning. Skilled in developing dashboards, writing SQL queries, and building end-to-end data pipelines that support business and IT decision-making. Proficient in working with structured and unstructured data, and applying NLP, statistical analysis, and system monitoring insights to drive data-informed outcomes.

Education

Master of Science, Data Science

University of North Texas – Denton, TX, USA

Expected May 2026

GPA: 4/4

Bachelor of Technology, Electrical Engineering

College of Technology and Engineering (CTAE) – Udaipur, Rajasthan, India

June 2023

GPA: 3.64/4

Technical Skills

- **AI & LLMs:** LLM Reasoning, Prompt Engineering, Fine-tuning, LangChain, Agentic Workflows, RAG, Transformers, Diffusion Models, Reinforcement Learning with Human Feedback (RLHF)
- **ML & Deep Learning:** Computer Vision, Object Detection (YOLOv5, Faster R-CNN), Gesture Analysis, Motion Modeling, Transformers, CNNs, GANs, Deep Learning (PyTorch, Keras/TensorFlow)
- **Data Analysis:** Exploratory data analysis, Data Cleaning, Feature Engineering, Microsoft Excel, Power BI, Tableau
- **Programming Languages:** Python, SQL, R, MATLAB, C++, Spark (PySpark)
- **Cloud Platforms:** Google BigQuery (ETL scripting, scheduled queries), AWS EMR (intro), S3, EC2, Lambda
- **Certifications:** Google Cloud - Performing Foundational Data ML and AI Tasks, Generative AI for Beginners by Microsoft, Generative AI with Large Language Models (Coursera)

Professional Experience

Graduate Research Assistant

University of North Texas – Denton, Texas

Sept 2024 – May 2025

- Fine-tuned Large Language Models (LLMs), integrated Retrieval-Augmented Generation (RAG) framework for efficient information retrieval, applied machine learning algorithms to analyze large datasets, enhancing the accuracy and efficiency of data processing.
- Evaluated summaries generated using diverse datasets and multiple evaluation metrics to validate the quality and reliability of LLM outputs and corresponding evaluation methods.

Data Analyst

MapUp India Pvt Ltd

Feb 2024 – June 2024

- Updated and maintained tolling systems on maps to ensure accurate representation of toll routes.
- Analyzed customer toll invoices to identify trends, provided actionable insights for cost reduction.
- Automated script execution with AWS Lambda, scheduling updates to ensure timely database synchronization.

Graduate Engineer Trainee

Secure Meters – Udaipur, Rajasthan, India

Jan 2023 – Feb 2024

- Developed and implemented a time-series prediction model, improving accuracy by incorporating business logic, achieved 90% accuracy for two business scenarios.
- Developed interactive data visualization dashboards using Plotly Dash, cleaned, processed, and validated data to ensure accuracy, enabling better data insights.

Project Experience

Motion-Controlled AI Agent (Prototype)

- Designed a lightweight system simulating human-like gesture and head movement using OpenPose-based skeleton tracking and CNN-RNN pipeline.

PatentAI: LLM-Based Patent Summarization & Retrieval

- Built an LLM-powered (Phi-4) system for summarizing patent claims and retrieving semantically similar documents using RAG, achieving 95% match accuracy in novelty detection and BERTScore > 0.85 for summarization quality.

Natural Language Route Planner

- Designed an agentic AI travel planner that interprets natural language instructions to generate dynamic, constraint-aware routes using NLP and OpenRouteService, with intelligent fallback and real-time navigation logic.

Deep Learning-Based Sketch-to-Art Generator

- Built a deep learning system to convert hand-drawn sketches into artwork using a U-Net and GAN pipeline, trained on the Sketchy dataset with data augmentation, real-time input.